GG-P-00661B(GSA-FSS)

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INTERIM REVISION OF
Fed. Spec. GG-P-661a
June 1, 1960

INTERIM FEDERAL SPECIFICATION

PROJECTOR, STILL PICTURE, OVERHEAD; ATTACHMENTS AND ACCESSORIES

This Interim Federal Specification was developed by the Standardization Division, General Services Administration, Federal Supply Service, Washington, D. C., 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification GG-P-661A, dated June 1, 1960.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers overhead projectors which project an image upon a vertical screen in a room lighted with normal intensity.

1.2 Classification.

1.2.1 Types and classes. Overhead projectors shall be of the following types and classes, as specified (see 6.2):

Types:

Type I. Standard.
Type II. Glare Reducing.

Classes:

Class A. Auditorium Type Projectors (see table I).
Class B. Classroom Type Projectors (see table I).
Class C. Collapsible Desk-Top Projectors with Integral Carrying Case (see table I).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids,or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

J-C-175 - Cable Assembly, Power, Electrical.
CC-M-636 - Motors, Fractional-Horsepower, Alternating Current.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

FSC 6730

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Method Of.

MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment.

MIL-G-174 - Glass, Optical.

MIL-C-675 - Coatings of Glass Optical Elements.

MIL-I-983 - Interior Communications Equipment, Naval Shipboard.

MIL-E-17555 - Preparation for Delivery of Electronic Equipment and Miscellaneous Electrical Equipment.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-150 - Photographic Lenses.

MIL-STD-169 - Extreme Temperature Cycle.

MIL-STD-170 - Moisture Resistance Test Cycle for Ground Signal Equipment.

MIL-STD-252 - Wired Equipment, Classification of Visual and Mechanical Defects for.

MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements For Equipment

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids, or request for proposal shall apply:

Underwriters' Laboratories (UL), Inc. Standards:

No. 73 - Motor Operated Appliances.
No. 114 - Office Appliances and Business Equipment.
No. 486 - Wire Connectors and Soldering Lugs.
No. 796 - Printed Wiring Boards.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, Long Island, New York 11746; or 1655 Scott Blvd., Santa Clara, California 95050.)

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American National Standards Institute (ANSI), Inc. Standard:

S1.4-1961- General Purpose Sound Level Meters, Specifications for.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y., 10018.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 Each contractor shall submit to the contracting agency proof that the projector he proposes to supply under this specification conforms to the requirements of the Underwriters' Laboratories, Inc., UL73 and 114, as applicable. The label, or listing with reexamination, of the UL, may be accepted as evidence that the projector conforms to this requirement.

3.1.2 In lieu of the label, or listing with reexamination, the contractor may submit independent proof, satisfactory to the contracting agency, that the projector conforms to the applicable requirements of the published standards including methods of tests of the UL73 and 114 standards.

3.1.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to assure acceptance of his material.

3.2 Preproduction sample. Unless otherwise specified (see 6.2), before production is commenced, a complete unit covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.4. Approval of the preproduction sample authorizes commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items. The approved preproduction model shall be used as a guide to inspect production units and may be presented for acceptance as the last unit of the contract.

3.3 Material. Material shall be as specified herein. Where material is not specified, a material shall be used which will meet the performance requirements of this specification.

3.3.1 Metal. Metals used shall be either corrosion resistant or suitably treated to resist corrosion in atmospheric conditions to which the projector will be subjected when in storage or during use.

3.3.2 Dissimilar metals. Dissimilar metals, especially brass, copper or steel in contact with magnesium, aluminum or other alloys, shall be avoided wherever practicable. When such contacts are unavoidable, the metal shall be properly protected against electrolytic corrosion. An interposing material compatible to each of the dissimilar metals or other suitable means shall be utilized to properly protect metal against electrolytic corrosion. No insulating material will be required between corrosion-resistant steel inserts and aluminum castings when the inserts are integrally cast in the aluminum. The metal referred to shall be the metal on the surfaces of the part; i.e., zinc shall include all zinc parts such as castings as well as zinc-coated parts, whether the zinc is electrodeposited or applied by hot dipping or by metal spraying, over similar or dissimilar metal parts. These requirements shall not apply to standard component parts of assemblies such as rivets, bolts, nuts, and washers which will be painted prior to being placed in service.

3.4 Cleaning.

3.4.1 Parts. After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.4.2 Units. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

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3.5 Design. The projector shall be equipped with a light source, blower cooling system, horizontal projection stage, a special optical system, and associated equipment such as a carrying case. The projector shall meet the requirements of table I.

TABLE I. Projector requirements

Projection area	Maximum projector weight	Lens focal length ± 1/2 inch	Minimum resolving power (3.12.1) Lines per millimeter		Focusing range inches square	Minimum screen lumens output (3.12.6) 1/	Minimum uniformity of screen illumination (3.12.1.3)
			Center	4 inches from center			
	Pounds	Inches					Percent
Type I and Type II							
Class A	40	14	20	5	45-204	3000	50
Class B	30	14	12	5	50-120	2000	30
Class C	20	10	5	5	50-100	1500	30

1/ Screen lumens to be calculated on a standard 60 inch square test target (see 4.4.12).

3.5.1 Projection. The projector shall be capable of projecting sharp images onto a vertical screen from transparencies, ranging in sizes up to 10 by 10 inches, when placed upon the horizontal projection stage. The image reflected upon the screen shall be clear, bright, and of high resolution.

3.5.2 Physical stability. The projector, in ready-for-operation position, shall not fall over when tipped in accordance with 4.4.2.

3.6 Reliability.

3.6.1 Rough handling. The projector shall be capable of withstanding rough handling without adversely affecting the projected image when tested in accordance with 4.4.23.

3.6.2 Climatic conditions. The projector shall be capable of operating under variable climatic conditions found in a classroom.

3.7 Maintainability.

3.7.1 Internal components. Internal components of the projector that require frequent cleaning shall be readily accessible without disassembly and without the use of special tools. The lamp shall be replaceable without removal of components or use of tools.

3.8 Securing of parts. Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the items in which they are used are subjected to specified bounce and shock tests. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements. When practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such a manner that failure of a single rivet or screw will not free the part completely.

3.9 Protection against fungus growth. When specified (see 6.2), materials used shall be fungus-resistant or suitably treated to resist fungus. When material susceptible to fungus is used, it shall be treated in accordance with MIL-T-152. This treatment shall in no way alter the proper functioning of the projector.

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3.10 Projector housing. The projector housing shall contain the light source, cooling system, reflectors, condensing lenses and mirror system required to direct the light through the transparency.

3.10.1 Doors. All doors shall be hinged and fastened by means of captive fasteners, and construction of the housing shall be such as to reduce the light spill to a minimum to where it will not be distracting to the audience.

3.10.2 Lamp and lens accesibility. The projector lamp shall be readily accessible for easy replacement. All surfaces of the condenser lenses and the front surface of the external reflector behind the lamp shall be accessible for cleaning without the use of tools other than a screwdriver. Lens, reflector, and mirror mounts shall be so constructed as to insure the accurate return of lenses and mirrors to their original position when they are replaced after having been removed for cleaning.

3.11 Electrical system.

3.11.1 Input voltage. The projector shall be capable of operating from a 115 volts ± 10 volts and 60 Hz per second ± 5 percent AC source.

3.11.2 Circuit control. The electrical circuits of the projection lamp and cooling system motor shall be so interlocked that the projection lamp cannot be turned on without activating either the motor or a temperature controlled timer for the motor.

3.11.3 Power cable assembly. The power cable assembly shall be in accordance with J-C-175. The cable used for electrical connections between the source of electrical power and the projector shall be three conductor standard flexible cord CO-03HOF3/16. (The green conductor to be used as the "grounding conductor".) The cable shall be 15 feet ± 1 inch long with one end permanently connected to the projector and strain relieved. The cable jacket shall be either black or gray.

3.11.4 Fusing. When specified (see 6.2), both "hot" (ungrounded) sides of the power supply line shall be properly fused. Fuses shall conform to MIL-I-983. All other units are to have fusing in one "hot" (ungrounded) side of the power supply line or a UL approved internally fused lamp in lieu of a line fuse. If a fused lamp is used, the projector shall be conspicuously and permanently marked with a warning notice to replace with the specific lamp or its equivalent.

3.11.5 Cooling system. The cooling system shall be either a motor driven centrifugal blower or fan, capable of cooling and ventilating the projector housing to meet the requirements of 3.19. The hot air blown from the projector housing shall be directed away from the operator. The vibration transmitted to the optical system shall not be of sufficient magnitude to visibly affect the projected image.

3.11.5.1 Cooling motor. The motor for driving the ventilating fan or blower shall be in accordance with CC-M-636 and shall be of sufficient size and capacity to properly operate the fan or blower. The motor shall be supplied with a built-in lubrication system and capable of 2000 hours of continuous operation.

3.11.6 Wiring. Wiring shall be neat and properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wires. Clearance between wiring and any parts which dissipate heat shall be provided. Wires in a continuous run between two terminals shall not be spliced. The wiring shall be in accordance with the applicable paragraphs of UL73 and UL114.

3.11.6.1 Connectors. The wiring connectors shall be in accordance with the applicable paragraphs of UL486.

3.11.7 Printed wiring boards. The printed wiring boards if used shall be in accordance with the applicable paragraphs of UL796.

3.11.8 Light source. The light source of the projector shall be a form of Tungsten Halogen lamp having a minimum rated life of 75 hours. The lamp shall have a rated voltage of 120 volts.

3.11.9 Switch. A two-position or three-position switch for turning the projector "ON" and "OFF" or "ON", "OFF" and "Blower" shall be located on the front panel of the projector. When specified (see 6.2), either a double pole, single throw or a four pole, double throw switch shall be used.

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3.11.10 Safety interlock. An electrical interlock shall be provided to automatically disconnect power to the lamp and blower by opening both sides of the power lines when either the lamp access door or other normally used entryway into the lamp and blower compartment for purposes of replacing or cleaning components is opened or removed.

3.12 Optical system.

3.12.1 Projection lens. The lenses shall be in accordance with the requirements in table I when tested in accordance with 4.4.13. The lenses shall be type VII MIL-STD-150. All air-to-glass surfaces of the lens shall be treated in accordance with MIL-C-675.

3.12.1.1 Projected image. The image produced by the projection lens shall be visually free from haze due to spherical aberration, internal barrel reflections, surface defects, glass defects and from both longitudinal and lateral chromatic aberration, when viewed from a distance of 10 feet. The allowable tolernace for one side of the zonal region shall be no more than 2 percent of the width of the projected picture area.

3.12.1.2 Distortion. The projected image of straight lines forming the sides of an 8 inch square, placed centrally over the projecting stage, shall not deviate more than one percent of their length from true straight lines (see 4.4.4).

3.12.1.3 Uniformity of screen illumination. Illumination on the screen shall be free from objectionable bands or patches differing in color or brightness from the adjacent parts of the screen. The average illumination at the four corners shall not be less than the percent of the illumination at the center of screen specified in table I (see 4.4.16).

3.12.2 Mirror, projection head. The projection head mirror shall be a front reflecting surface mirror fabricated of glass in accordance with MIL-G-174. Any projection head mirrors that are not completely protected from dust and are open to the air shall have the reflecting surfaces protected from scratches and reflection losses in accordance with MIL-C-675.

3.12.3 Reflector, projection lamp. The concave projection lamp reflector shall be of first quality material throughout and shall not crack, flake, peel, discolor, or otherwise suffer impairment in quality of performance when subjected to 8 hours of operation in the projector housing or the moisture resistance test of 4.4.6.

3.12.4 Condenser lenses.

3.12.4.1 Glass lenses. If an optical glass condensing lens system is supplied with the equipment, the condensing lenses shall be type XI in accordance with the requirements of MIL-STD-150.

3.12.4.2 Plastic lenses. If a plastic Fresnel type condensing lens system is supplied with the equipment, the plastic condensing lens shall be capable of withstanding the heat produced by the projection lamp without causing any defect to the lens which would impair the quality of the projected picture, when projecting a transparency having a uniform density of 2.0 for a continuous operational period of 4 hours.

3.12.5 Heat absorbing filter. For transparency protection, a heat absorbing filter, for lamps producing over 2600 lumens on the screen shall be installed between the lamp and projector stage. The efficiency of light transmission of the filter shall be at least 85 percent and the total infrared transmission shall not exceed 4 percent.

3.12.6 Light output. The light output of the projector with the lens system in focus and operating at rated voltage shall be as specified in table I when measured in accordance with 4.4.12.

3.12.7 Glare reducer. When specified (see 6.2), the projector shall contain a glare reducing capability designed so that the glare to the eyes of any operator ranging in height from 5' -0" to 6' -4" and standing in either a right hand or left hand operating position shall be no more than 35,000 foot lamberts as measured in 4.5. Any glare reducing device shall not interfere with the operator in the normal performance of his duties.

3.13 Projection system.

3.13.1 Projection system. A dustproof projection head firmly positioned and aligned with the optical axis, shall be provided.

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3.13.2 Projection head support. The support arm connecting the projection head to the vertical post shall be of sufficient strength to support the projector without permanent deflection taking place when the projector is tested in accordance with 4.4.6. When the projector head is connected directly to the vertical post by either forming or bending the vertical post, a support arm is not required. The horizontal and vertical supports shall be so located so as not to interfere with the complete observance of the transparency.

3.13.3 Elevating device. Means shall be provided for raising the image to 20° above the horizontal. Once set, the position of the image shall remain fixed until changed by hand (see 4.4.7). An adjustment shall be provided for increasing the tension on the elevating part of the projector head to assure that the position will remain as set even after usage.

3.13.4 Focusing mechanism. An easily operable rack and pinion free from detrimental backlash or a nonslip friction focusing mechanism, attached to the projection head support, shall be provided, which will produce a clear steady focus of the image in accordance with table I. Once the focus is set, it shall remain in position without creep until changed by hand. An adjustment shall be provided on the friction type focusing mechanisms for increasing the amount of friction to compensate for wear.

3.14 Projector feet. Each corner of the projector shall be provided with a captive, nonmarking, foot to prevent sliding of the projector and to prevent damaging of the table top surface.

3.15 Flaking. Lacquers or blackening inside the projection head used to deaden reflections shall be free from dusting or flaking resulting from exposure to any test outlined in this specification.

3.16 Projection stage. A projection stage approximately 10 inches by 10 inch in size, vented to insure dissipation of heat, shall be provided. The top surface shall contain a glass plate, meeting the requirements of MIL-G-174. It shall be suitable for projecting written material, and for the projection of transparencies placed thereon either in a lengthwise or crosswise position. The stage shall be so constructed as to accept a roll attachment accessory operating from either top to bottom or side to side. The fresnel lens shall be readily accessible for cleaning without the use of tools other than a screwdriver. The glass plate shall be easily removable without the use of tools other than a screwdriver.

3.16.1 Reversible roll mechanism transparency. When specified (see 6.2), the projector shall be equipped with two (2) roll attachments capable of accepting 50 feet of 10-1/2 ± 1/4 inch wide polyester or acetate. The material shall pass directly over the projection stage onto the other roll attachment. The roll attachments shall be provided with means for advancing and returning the strip. One of the two (2) roll attachments shall be furnished with 50 feet either polyester or acetate.

3.16.1.1 Scratch film. When plastic strip is rolled per paragraph 3.16.1, there shall be no evidence of any scratches to film surface (see 4.3.6).

3.17 Heavy duty carrying case. When specified (see 6.2), the projector carrying case shall be of sturdy construction to provide the contents protection from weather, blows, and impact during transportation and handling. The case shall meet the requirements specified in paragraph 3.22.6. If the combined weight of the projector and case exceeds 45 pounds, the carrying case shall be supplied with four replaceable casters, one located on each corner of the underside of the case. Two of the casters shall be of the swivel type. The casters shall permit rolling along the floor or deck. The case shall be of such size as to contain the projector plus accessories except stand.

3.17.1 Carrying handles. The carrying case shall contain three handles as follows: One handle located on the top essentially above the center of gravity of the case when packed for carrying and parallel to the longest side, and one on each end of the case to facilitate two-man carry.

3.17.2 Corner reinforcements. Copper-nickel alloy or brass corner reinforcements shall be provided on the carrying case and shall be so designed to permit sliding along the floor and to permit stacking for storage.

3.17.3 The combined weight of the projector, case, and associated equipment shall not exceed 75 pounds (see 4.4.3).

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3.18 Light duty carrying case. Unless otherwise specified (see 6.2), the projector shall be equipped with a light duty carrying case which shall be a molded, light weight, trunk type enclosure designed to house the projector and component parts. The case may be either completely detachable from the projector or utilize the projector base as an intregal part of the projector case. A handle shall be permanently secured on the top of the case and of sufficient strength to prevent breaking or loosening from the case under normal carrying conditions.

3.19 Temperature rise. The temperature rise of the projector housing, after 2 hours of continuous operation shall not exceed 50°F. above an ambient temperature of 70°F. exclusive of the blower exhaust, and no damage shall result to the transparency roll (see 4.4.15).

3.20 Noise of cooling system. The noise of the cooling system when measured as described in 4.4.10, shall not be greater than 46 decibels above a reference level of ten to the minus 16 watts per square centimeter in accordance with ANSI - S1.4-1961.

3.21 Life test. With exception of the projector lamp there shall be no excessive wear or damage to any part of the projector when projector is operated in accordance with 4.4.8.

3.22 Service conditions. When specified (see 6.2), the equipment shall meet the following service conditions; when a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.22.1 Temperature. When tested in accordance with 4.4.14, the performance shall be as follows:

(a) Operating: Ambient temperature in the range of 40° to 105°F.; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature not to exceed 72 hours, at any one time.

(b) Nonoperating: Ambient temperature in the range of -40°F. to +150°F.; exposure at high temperature extreme not to exceed 4 hours, at a low temperature not to exceed 24 hours, at any one time.

3.22.2 Relative humidity. The projector shall meet all operating requirements of this specification, when tested in accordance with 4.4.9.

3.22.3 Drop requirements. When packed in the carrying case with the projection lamp removed, the equipment shall not be damaged nor the function of the equipment impaired when tested as specified in 4.4.5.

3.22.4 Vibration, equipment. The natural frequency of the equipment shall be between 25 and 35 cycles per second. The amplitude or vibration of the mounted equipment shall not exceed three times the amplitude of the applied vibration at any frequency between 10 and 25, or between 25 and 55, cycles per second (see 4.4.17).

3.22.5 Vibration, shipboard. The design of the projector shall be such that use under shipboard conditions of underway vibration will not appreciably affect the projected image.

3.22.6 Bounce. The equipment after subjection to test in 4.4.22 shall meet full specification performance. There shall be no physical damage except abrasion.

3.22.7 Preconditioning. The equipment shall be capable of meeting the requirements specified herein without processing, after subjection to the bounce preconditioning specified in paragraph 4.4.21.

3.22.8 Radio-interfernce suppression. The radio-interference suppression requirements for the projector shall conform to MIL-STD-461.

3.23 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment (see 4.3.5). Individual items shall not be matched for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.24 Technical literature and running spare parts. When specified (see 6.2), technical literature and running spare parts shall be furnished by the contractor. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.

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3.25 Operational. When tested, as specified in 4.4.11, it shall be determined that the following requirements specified herein are complied with:

(a) Blower motor and lamp operation in accordance with 3.11.2.

(b) The projected image in accordance with 3.12.1.1 and 3.12.1.3.

3.26 Workmanship. The item shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.4 Cleaning.

3.8 Securing of parts.

3.11 Electrical Systems.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection lot. Unless otherwise specified (see 6.2), all projectors of the same type and size (including carrying case and accessories) presented for delivery at one time shall be considered a lot for purposes of acceptance inspection.

4.3 Inspections.

4.3.1 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery (see 4.3.2).

(b) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.3.3).

(2) Quality conformance inspection of preparation for delivery (see 4.3.4).

4.3.2 Preproduction inspection. This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table II, and the inspection specified for group A, Group B, and group C (see tables III, IV, and V. respectively).

TABLE II. Preproduction inspection

Inspection (For additional preproduction inspection, see 4.3.2)	Req. Para.	Test Para.
Bounce test.	3.22.6	4.4.22
Moisture resistance.	3.22.2	4.4.9
Temperature.	3.22.1	4.4.14
Vibration.	3.22.4	4.4.17
Weight.	3.17.3	4.4.3

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4.3.3 Inspection of equipment prior to delivery. The contractor shall perform the inspection specified in 4.3.3.1 through 4.3.3.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspections to verify the contractor's compliance with requirements. Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A, group B, or group C inspection, except preproduction samples, shall be preconditioned after final assembly.

4.3.3.1 Group A inspection. This inspection, including sampling, shall conform to table III and the normal inspection procedures of MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE III. Group A inspection

Inspection	Req. Para.	Test Para	AQL	
			Major	Minor
Visual and mechanical.	3.26	4.3.6	1.5 dphu	6.5 dphu
Optical				
Distortion.	3.12.1.2	4.4.4)		
Elevation of projected image.	3.13.3	4.4.7)		
Screen lumens output.	3.12.6	4.4.12)	1.5 dphu combined	*
Resolving power.	3.12.1	4.4.13)		
Uniformity of screen illumination	3.12.1.3	4.4.16)		
Operational.	3.25	4.4.11	1.0 percent	*

*All optical and operational defects are considered major.

4.3.3.2 Group B inspection. This inspection, including sampling, shall conform to table IV and to the special procedures for small-sample inspection of MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and meet the group A inspection. The AQL shall be 6.5 percent defective and the inspection level shall be S-4.

TABLE IV. Group B inspection

Inspection	Req. Para.	Test Para.
Interchangeability	3.23	4.3.5
Noise	3.20	4.4.10

4.3.3.2.1 Order of inspection within group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

4.3.3.3 Group C inspection. This inspection shall be as listed in table V and shall normally be performed on sample units that have been subjected to and meet group A and group B inspection.

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4.3.3.3.1 Sampling for group C inspection. Samples shall be selected in accordance with the following:

(a) One unit from each 100 or fraction thereof produced on order for temperature rise test.

(b) One unit from each 250 or fraction thereof produced on order for life test.

4.3.3.3.2 Noncompliance. If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

TABLE V Group C inspection

Inspection	Req. Para.	Test Para.
Temperature rise	3.19	4.4.15
Life	3.21	4.4.8
Drop test	3.22.3	4.4.5

4.3.3.4 Reinspection of conforming group B and group C sample units. Unless otherwise specified (see 6.2), sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.3.4 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

4.3.5 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.23. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(c) Location of locking pins, fasteners, slides and mountings which receive mating parts.

(d) Dimensions common to projector housing, projector head support, and projector head which govern mounting and alignment.

4.3.6 Visual and mechanical inspection. Equipment shall be examined for the defects listed in MIL-STD-252 and table VI.

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TABLE VI. Classification of visual and mechanical defects

Classification	Defects
Major	1. Focusing mechanism binds, sticks or freezes. 2. Projection head will not remain in operating position. 3. Parts missing or broken. 4. Broken or defective controls, binds, sticks or freezes. 5. Lens chipped or cracked. 6. Hinge or fastener missing. 7. Film scratched, roll mechanism operation.
Minor	1. Input power cable not strain relieved. 2. Input power cable has wrong dimensions. 3. Weight exceeds specified limit. 4. Fasteners do not fasten securely. 5. Doors do not fit properly. 6. Finish not as specified. 7. Handles on case not properly located. 8. Slight damage or dents in case.

4.4 Tests. The following tests shall be conducted in order to determine whether or not the projector conforms to the requirements of this specification:

4.4.1 Test conditions. The projector shall be tested with 118-volt, 60-Hertz alternating current. The tolerance for the standard or otherwise indicated voltage and frequency shall be ± 2 percent in voltage and ± 5 percent in frequency. The ambient temperature shall be 70° ± 5°F.

4.4.1.1 Operating voltage. The projector shall be operated continuously for one hour at 125-volts, 60-Hertz AC and one hour at 105-volts, 60-Hertz AC to test for compliance with paragraph 3.11.1.

4.4.2 Physical stability. The projector, ready for operation with the head racked up to its maximum elevation shall be tipped 12° from the vertical in each of four directions to determine compliance with 3.5.2. The projector shall then be operated at angles of 10° for a period of 6 hours without operational failure. Such operation shall include using the roll mechanism, focusing, positioning transparencies and the like.

4.4.3 Weight. The equipment shall be weighed to determine compliance with 3.17.3.

4.4.4 Distortion. The lens under test shall be used to project an image of a test plate having straight lines adjacent to the edges of the picture aperture. The projected image shall be on a white matte grainless screen located at such a distance from the projector that each side of the projected image of the test plate shall be approximately 72 inches. The straight lines drawn immediately adjacent to the edges of the picture aperture shall be observed on the screen and shall not be distorted more than specified in 3.12.1.2. Only one lens per lot shall be tested.

4.4.5 Drop. The equipment, packed for carrying shall be dropped ten times with the base of the carrying case in a plane parallel with the floor from a height of 12 inches. The case and equipment shall than be examined for loosening of nuts, bolts, screws, breakage of mirrors and other temporary or permanent damage to determine conformance with 3.22.3. Minor case deformation shall not be cause for rejection.

4.4.6 Support arm deflection. A clamp or similar holding device shall be securely fastened to the projection head support arm when the projector is assembled and ready for operation. Attach one end of a chain to the clamp and the other to a rigid support. Allowing the projector to hang free supported only by the clamp, raise the projector by the base or sides approximately six inches and release allowing the projector to drop free until stopped by the chain and clamp. Repeat the drop six times. Check support arm for conformance with 3.13.1 and 3.13.2.

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4.4.7 Elevating projected image. The projected image shall be moved through an arc of 20° to determine compliance with 3.13.3.

4.4.8 Life. The equipment shall be operated continuously for 50 hours at 70° ± 5° Ambient Temperature to determine compliance with the requirement specified in 3.21.

4.4.9 Moisture-resistance test for equipment. The projector set-up in normal operating position shall be tested as follows:

4.4.9.1 Test method.

(a) Dry at 130°F. for 24 hours.
(b) Condition at 77° ± 5°F. and 40 to 50 percent relative humidity for 24 hours.
(c) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-17C.

(d) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5°F. and 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. The equipment shall meet full specified performance for those measurements specified in 4.4.9.

4.4.9.2 Failure. If the equipment fails to meet the full specification requirements, as specified in 4.4.9.1(d), it does not pass the test.

4.4.10 Noise of cooling system. The projector shall be mounted so that it can be turned on a vertical axis through the picture aperture. The microphone of a sound-level meter, in accordance with the requirements of ANSI S1.4-1961 shall be placed 5 feet from this vertical axis and in a horizontal plane passing through the picture aperture. Readings of the sound level shall be taken at 45 degree intervals beginning with a reference position in front of the projector and on a horizontal level of the picture aperture while the projector is operating and with the meter at the curve B setting (sec 3.20).

4.4.11 Operational test. The projector shall be operated with a transparency having a density reading of 1. The image shall be inspected and the controls shall be operated to determine compliance with 3.25.

4.4.12 Screen lumens output. To determine the screen lumens output. The projector shall be focused on a screen in a plane perpendicular to the optical axis of the projector. The screen illuminated area shall be 5 feet square and shall be divided into 25 equal squares. The direct (not reflected) illumination at the center of each square shall be measured with a photoelectric light meter corrected by suitable color filters to conform to within a good approximation to the visibility curve of the human eye. The sum total of the 25 illumination measurements in foot-candles equals the light output in lumens. All lenses shall be clean and in proper adjustment when making the measurements.

4.4.13 Resolving power. The lens to be tested shall be mounted in a projector. This test shall be performed in accordance with MIL-STD-150, except as noted below:

5.1.2.12.4 - Method 15 Projected Visual Resolving Power.

A complete standard projector shall be used, projecting an image 60 inches wide. The test transparency shall be of photographic or a diazo type. A polished glass plate shall be used to hold the transparency flat on the projection stage. Test pattern groups shall be located as follows: One test group located in the center of the projection stage on the optical axis and eight groups located on the circumference of an 8-inch diameter circle at 45° intervals.

4.14.14 Temperature test. The equipment shall be subjected to the temperature cycle shown in MIL-STD-169, except the steps shall be as indicated.

Step	Measurement	Performance
1, 3, 4, 5, 8, 9 and 10	Operation	Full specification requirements

4.14.15 Temperature rise. The equipment shall be operated and tested to determine compliance with the requirement specified in 3.19. The instrument used to determine the rise in temperature shall be accurate to ± 5°F. The temperature rise at the hottest spot of the projector housing shall be measured. A test transparency of uniform density 2.0 or greater shall then be placed on the projection stage, and the temperature of the transparency measured at the end of 5 minutes to determine conformance to 3.19.

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4.4.16 Uniformity of screen illumination. The uniformity of screen illumination shall be determined using the same location and test conditions specified in 4.4.12 for compliance with 3.12.1.3.

4.4.17 Test for equipment vibration. The equipment shall be tested as follows, to determine conformance to 3.22.4.

(a) Secure the equipment to a vibration table that can be controlled within 10 percent of the specified amplitude. The equipment shall be mounted in its shipping container.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be 0.030 inch; however, this may be reduced, if necessary, to prevent excessive excursion at or near resonance provided that the excursion can still be measured.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure amplitude of the applied vibration and the vibration of the equipment for frequencies between 10 and 25 and between 35 and 55 cycles per second.

(e) Determine the natural frequency of the equipment; the frequency at which the ratio of the equipment vibration to the applied vibration is greatest shall be considered the natural frequency.

4.4.18 Image vibration. During the resolution test specified in 4.4.1.3, the presence of image blurring caused by vibration shall be observed. Any such blurring seen from a distance of 18 inches from the screen shall be considered as unacceptable in accordance with 3.11.5

4.4.19 Color bands and patches. During the tests specified any presence of color bands, patches or interference patterns shall be noted. Any such phenomena visible at a distance of 10 feet or greater from the screen shall be cause for rejection.

4.4.20 Dielectric test. The insulation strength of all electrical circuits shall be tested to withstand twice the rated voltage plus 1,000 volts. The test potential may be direct current or peak of the same value and shall be applied for 1 minute between all current-carrying parts and adjacent conductors not in the circuit and between all wiring and ground, without puncturing of insulation or flashover occurring. The power source used during this test shall have a capacity of at least 1 kilowatt.

4.4.21 Bounce preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall operate at a speed to make possible the insertion of 1/32 inch strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for one minute. After bounce preconditioning, the unit shall not be repaired, cleaned, or otherwise changed prior to subjection to acccptance inspection.

4.4.22 Bounce test. The equipment shall be tested on the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The test shall be as follows:

(a) Cover the tester bed with a panel of 1/2 inch plywood, with the grain parallel to the drive chain. Space six-penny nails, with the heads driven below the surface, at 6 inch intervals and around all four edges and at a 3 inch intervals in a 6 inch square in the center.

(b) Place the equipment in its transit case, prepared for field transport, on the bed of the package tester. Limit the lateral motion by wooden fences to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the back-stop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 $\pm$ 2 rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

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4.4.23 Rough handling test (preparation for delivery). When rough handling test in accordance with MIL-P-116, paragraph 4.4.6 is required by the contract (see 6.2), the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling.

(a) Paragraph 4.3.6 Visual and mechanical.

(b) Paragraph 4.4.11 Operational test.

4.5 Procedure for measurement of glare. Turn on the projector and allow it to warm up. Mark on the projection stage a longitudinal centerline using the direction of the projected image as "0" degrees. Similarly mark on the stage a transverse centerline intersecting the longitudinal centerline at the center of the stage. The intersecting angle shall be 90°. Using a brightness spot meter and an adjustable stand, mount the meter 30 inches from the center of the stage and on the same plane as the stage. A line projected through the meter and along the line of sight of the spot shall intersect the junction of the two centerlines at an angle of 60°. Maintaining this same horizontal angle and distance, elevate the meter until the projected line of sight of the spot forms a vertical angle to the plane of the stage between 45 and 65°. This vertical angle is the viewing angle for different height operators (see 3.12.7). While moving the meter through an arc of 45 to 65°, focus the meter on the brightest spot on the projection stage. The foot lambert reading shall conform to 3.12.7.

4.6 Preservation, packaging, packing and marking. The inspector shall ascertain that the preservation, packaging, packing and marking of the projector and associated equipment conform to this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. The projector, attachments, and accessories shall be preserved and packaged by level A or C; packed by level A, B, or C, as specified (see 6.2), in accordance with the requirements for Projector, Opague in MIL-E-17555.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, the unit package and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking required by the contract or order, the unit package and shipping container shall be marked in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. Projector, still picture, overhead is intended for use by a teacher, lecturer, or speaker to project transparencies on a screen to be observed by an audience in a normally lighted room.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (1.2).
(c) Preproduction samples, if other than specified (3.2).
(d) Tropicalization of material, if required (3.9).
(e) Fusing, if other than specified (3.11.4).
(f) Special switch, if required (3.11.9).
(g) Glare reducer, if required (3.12.17).
(h) Roll mechanism, if required (3.16.1).
(i) Heavy duty carrying case, if required (3.17).
(j) Light duty carrying case unless otherwise specified (3.18).
(k) Service conditions, if required (3.22)
(l) Technical literature and running spare parts, if required (3.24).
(m) Responsibility for inspection, if other than specified (4.1).
(n) Inspection lot other than specified (4.2).
(o) Reinspection required unless otherwise specified (4.3.3.4).
(p) Rough handling test, if required (4.4.23).
(q) Level of packaging and packing required (5.1).
(r) Marking, if other than specified (5.2).

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